

Problem 3. Consider a parallelogram that has side lengths a and b with an angle of θ between them. The area of the parallelogram is $A = ab \sin \theta$.

- (a) Use the chain rule to write an approximate formula for the uncertainty ΔA in terms of the uncertainties Δa , Δb and $\Delta \theta$.

The chain rule says that

$$\begin{aligned}\Delta A &\approx \frac{\partial A}{\partial a} \Delta a + \frac{\partial A}{\partial b} \Delta b + \frac{\partial A}{\partial \theta} \Delta \theta \\ &= b \sin \theta \Delta a + a \sin \theta \Delta b + ab \cos \theta \Delta \theta.\end{aligned}$$

- (b) Suppose that you measure the quantities $a = 10$ cm, $b = 15$ cm and $\theta = \pi/6$ radians. If your measuring equipment has uncertainties $\Delta a = \Delta b = 0.1$ cm and $\Delta \theta = \pi/180$ radians, use your formula from part (a) to estimate the uncertainty ΔA . You can leave your answer in unsimplified form.

Substituting $a = 10$, $b = 15$, $\theta = \pi/6$, $\Delta a = \Delta b = 0.1$ and $\Delta \theta = \pi/180$ gives

$$\begin{aligned}\Delta A &\approx b \sin \theta \Delta a + a \sin \theta \Delta b + ab \cos \theta \Delta \theta \\ &= (15) \sin(\pi/6)(0.1) + (10) \sin(\pi/6)(0.1) + (10)(15) \cos(\pi/6)(\pi/180) \\ &= (15)(1/2)(0.1) + (10)(1/2)(0.1) + (10)(15)(\sqrt{3}/2)(\pi/180) \\ &\approx 3.5 \text{ cm}^2.\end{aligned}$$

Remark: The computed area is $A = (10)(15) \sin(\pi/6) = 75 \text{ cm}^2$, so the relative error is $\Delta A/A = 3.5/75 \approx 4.7\%$.

Problem 4. Consider the scalar field $f(x, y, z) = x^2 + y^4 + z^2 x$.

- (a) Compute the gradient vector field ∇f .

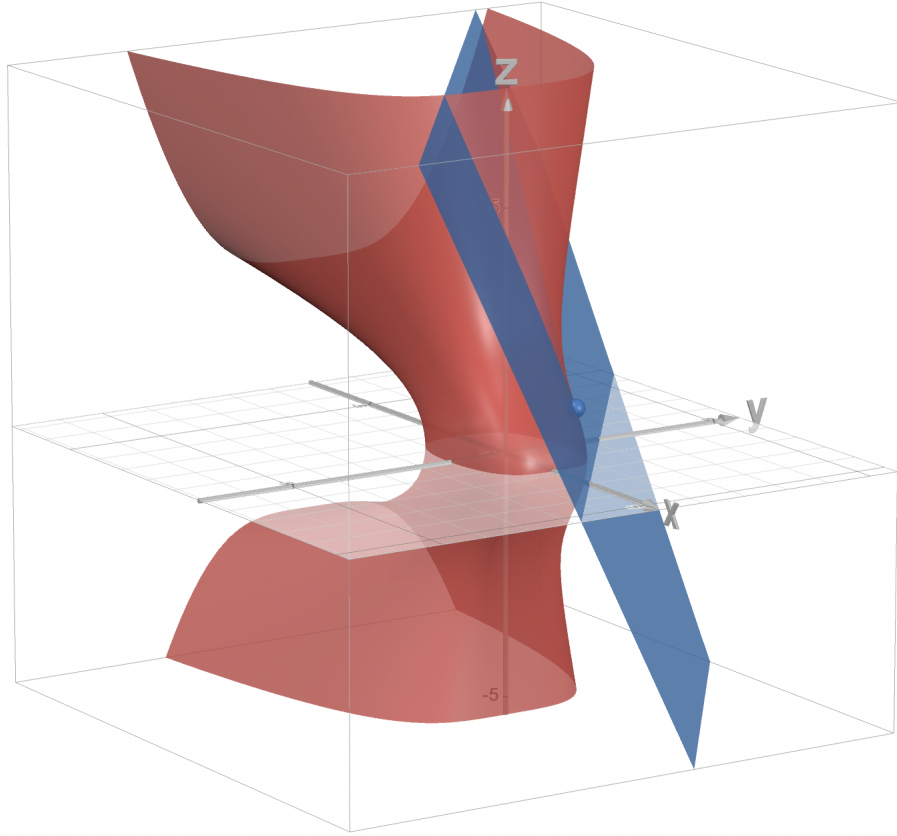
$$\begin{aligned}\nabla f(x, y, z) &= \langle f_x, f_y, f_z \rangle \\ &= \langle 2x + z^2, 4y^3, 2zx \rangle.\end{aligned}$$

- (b) Note that $f(1, 1, 1) = 3$. Use your answer from part (a) to find the equation of the tangent plane to the surface $x^2 + y^4 + z^2 x = 3$ at the point $(1, 1, 1)$.

The equation of the tangent plane to the level surface $f(x, y, z) = 3$ at the point $(1, 1, 1)$ is

$$\begin{aligned}\nabla f(1, 1, 1) \bullet \langle x - 1, y - 1, z - 1 \rangle &= 0 \\ \langle 2(1) + (1)^2, 3(1)^3, 2(1)(1) \rangle \bullet \langle x - 1, y - 1, z - 1 \rangle &= 0 \\ \langle 3, 4, 2 \rangle \bullet \langle x - 1, y - 1, z - 1 \rangle &= 0 \\ 3(x - 1) + 4(y - 1) + 2(z - 1) &= 0 \\ 3x + 4y + 2z &= 9.\end{aligned}$$

Here is a picture:



Problem 5. The function $f(x, y) = \sin(x) \sin(y)$ has infinitely many critical points. Here are three of them:

$$(0, 0), \quad (\pi/2, \pi/2), \quad (\pi/2, -\pi/2).$$

(a) Compute the Hessian matrix of $f(x, y)$ and its determinant.

First we compute all first and second derivatives:

$$\begin{aligned} f_x &= \cos(x) \sin(y), \\ f_y &= \sin(x) \cos(y), \\ f_{xx} &= -\sin(x) \sin(y), \\ f_{yy} &= -\sin(x) \sin(y), \\ f_{xy} &= \cos(x) \cos(y), \\ f_{yx} &= \cos(x) \cos(y). \end{aligned}$$

Hence the Hessian determinant is

$$\begin{aligned} \det(Hf) &= \det \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} \\ &= \det \begin{pmatrix} -\sin(x) \sin(y) & \cos(x) \cos(y) \\ \cos(x) \cos(y) & -\sin(x) \sin(y) \end{pmatrix} \\ &= [\sin(x) \sin(y)]^2 - [\cos(x) \cos(y)]. \end{aligned}$$

- (b) Use the second derivative test to determine whether each of the three critical points listed above is a local max, local min or a saddle point.

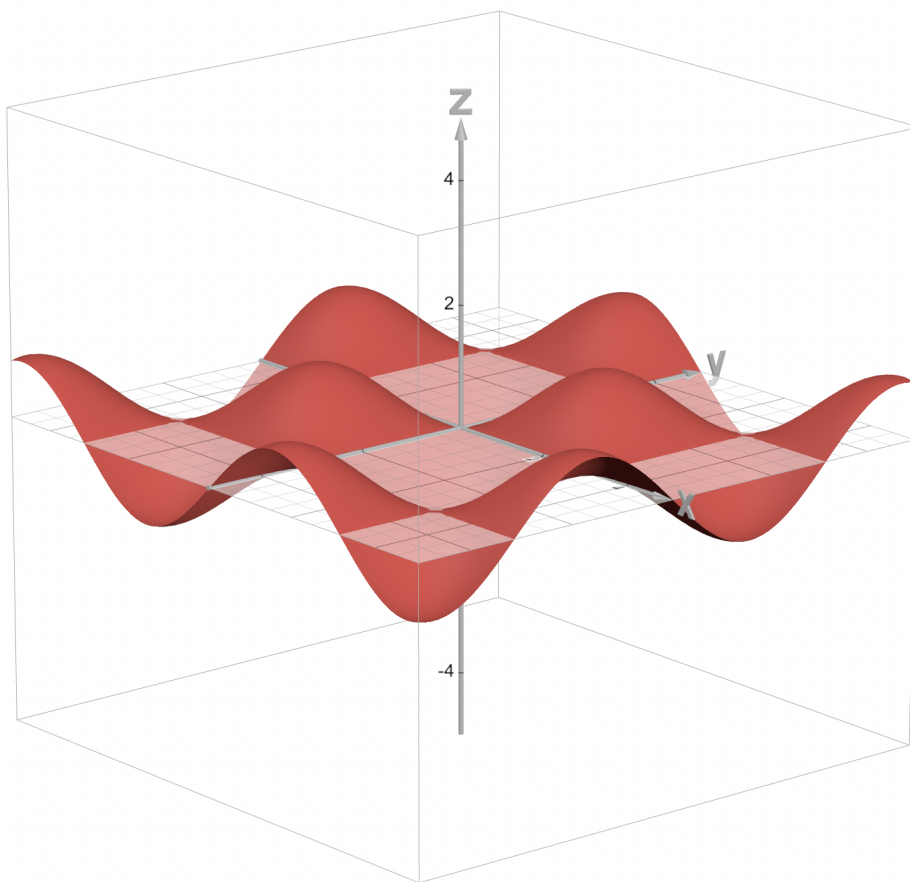
The critical point $(0, 0)$ satisfies $\det(Hf)(0, 0) = [\sin(0)^2]^2 - [\cos(0)^2]^2 = 0 - 1 = -1 < 0$, so this is a saddle point. The critical point $(\pi/2, \pi/2)$ satisfies

$$\det(Hf)(\pi/2, \pi/2) = [\sin(\pi/2)^2]^2 - [\cos(\pi/2)^2]^2 = 1^2 - 0^2 = 1 > 0,$$

so this is a local max or min. Since $f_{xx}(\pi/2, \pi/2) = -\sin(\pi/2)^2 = -1 < 0$ it is a local max. The critical point $(\pi/2, -\pi/2)$ satisfies

$$\begin{aligned} \det(Hf)(\pi/2, -\pi/2) &= [\sin(\pi/2) \sin(-\pi/2)]^2 - [\cos(\pi/2) \cos(-\pi/2)]^2 \\ &= [(1)(-1)]^2 - [0]^2 \\ &= 1 < 0, \end{aligned}$$

so this is a local max or min. Since $f_{xx}(\pi/2, -\pi/2) = -\sin(\pi/2) \sin(-\pi/2) = -(1)(-1) = 1 > 0$ it is a local min. Here is a picture:



Problem 1. Chain Rule. Let $f(x, y) = xe^y$ with $x(u, v) = uv$ and $y(u, v) = u^2v$. Use the multivariable chain rule to compute df/du .

The chain rule says that

$$\frac{df}{du} = \frac{df}{dx} \cdot \frac{dx}{du} + \frac{df}{dy} \cdot \frac{dy}{du}, \quad \text{or} \quad f_u = f_x \cdot x_u + f_y \cdot y_u.$$

In order to compute this we compute the partial derivatives:

$$\begin{aligned} f_x &= e^y, \\ f_y &= xe^y, \\ x_u &= v, \\ y_u &= 2uv. \end{aligned}$$

Then we put them together:

$$f_x = ve^y + 2uvxe^y.$$

If we want, we can express the answer in terms of u and v :¹

$$f_x = ve^{u^2v} + 2uv(uv)e^{u^2v}.$$

Problem 2. Tangent Plane. Let $f(x, y, z) = x^2y - z$. Find the equation of the tangent plane to the surface $f(x, y, z) = 1$ at the point $(x_0, y_0, z_0) = (2, 1, 3)$.

The equation of the tangent plane to the surface $f(x, y, z) = \text{constant}$ at a point (x_0, y_0, z_0) is

$$\begin{aligned} \nabla f(x_0, y_0, z_0) \bullet \langle x - x_0, y - y_0, z - z_0 \rangle &= 0, \\ \langle f_x(x_0, y_0, z_0), f_x(x_0, y_0, z_0), f_x(x_0, y_0, z_0) \rangle \bullet \langle x - x_0, y - y_0, z - z_0 \rangle &= 0, \\ f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0) &= 0. \end{aligned}$$

In our case we have

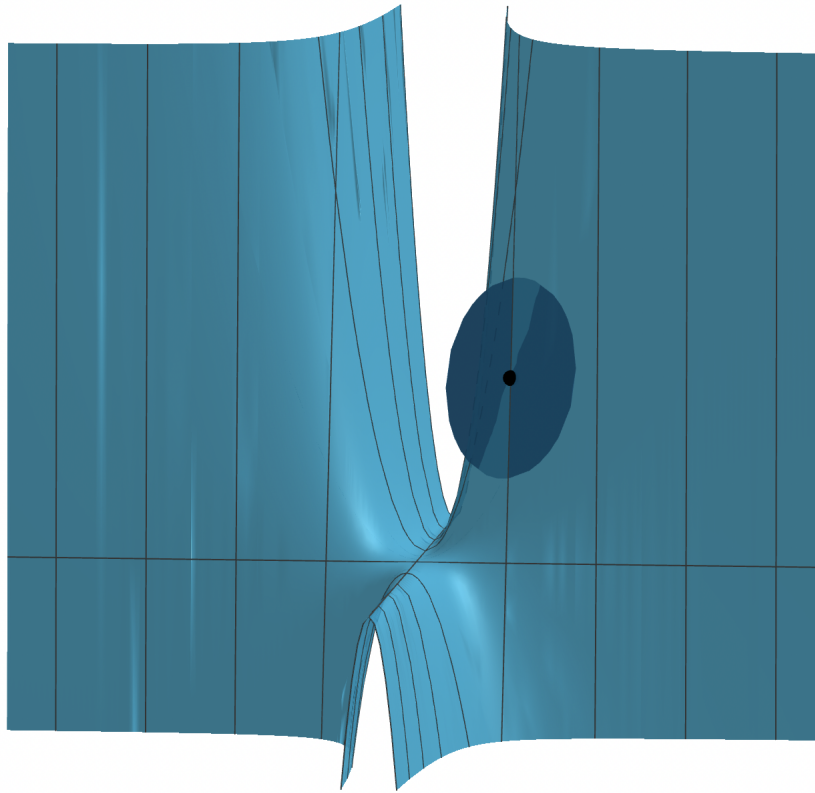
$$\begin{aligned} f_x &= 2xy, \\ f_y &= x^2, \\ f_z &= -1, \end{aligned}$$

and $(x_0, y_0, z_0) = (2, 1, 3)$, so the tangent plane is

$$\begin{aligned} f_x(2, 1, 3)(x - 2) + f_y(2, 1, 3)(y - 1) + f_z(2, 1, 3)(z - 3) &= 0 \\ 2(2)(1)(x - 2) + (2)^2(y - 1) - 1(z - 3) &= 0 \\ 4(x - 2) + 4(y - 1) - 1(z - 3) &= 0 \\ 4(x - 2) + 4(y - 1) - 1(z - 3) &= 0 \\ 4x + 4y - z &= 9. \end{aligned}$$

Here is a picture:

¹And, I guess, we could simplify it.



Problem 3. Optimization. The scalar field $f(x, y) = x^3 - y^3 + xy$ has two critical points: $(0, 0)$ and $(1/3, -1/3)$. Use the second derivative test to determine whether each of these is a local maximum or minimum, saddle point, or degenerate.

We need to compute the determinant of the Hessian matrix. We begin by computing the second partial derivatives:²

$$\begin{aligned} f_x &= 3x^2 + y, \\ f_y &= -3y^2 + x, \\ f_{xx} &= 6x, \\ f_{yy} &= -6y, \\ f_{xy} &= 1, \\ f_{yx} &= 1. \end{aligned}$$

Thus the Hessian determinant is

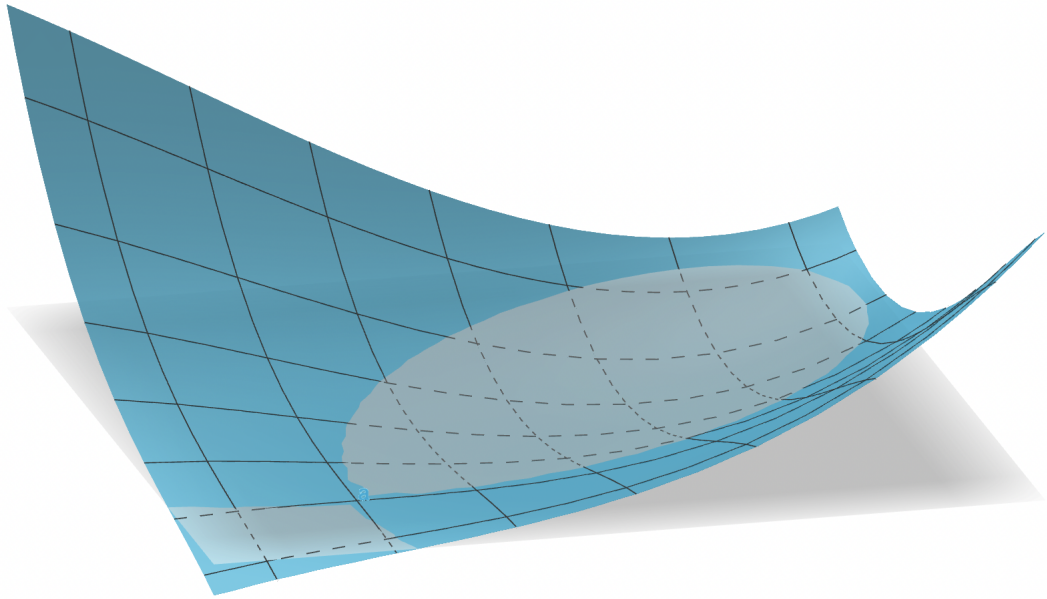
$$\det(Hf) = \det \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = \det \begin{pmatrix} 6x & 1 \\ 1 & -6y \end{pmatrix} = -36xy - 1.$$

Since $\det(Hf)(0, 0) = -1 < 0$ we see that $(0, 0)$ is a saddle point. Since $\det(Hf)(1/3, -1/3) = -36(-1/9) - 1 = 3 > 0$ we see that $(1/3, -1/3)$ is a local maximum or minimum. Since $f_{xx}(1/3, -1/3) = 6(1/3) = 2 > 0$ we see that $(1/3, -1/3)$ is a local minimum.³

²One can check that $\langle f_x(0, 0), f_y(0, 0) \rangle = \langle 0, 0 \rangle$ and $\langle f_x(1/3, -1/3), f_y(1/3, -1/3) \rangle = \langle 0, 0 \rangle$, as claimed.

³We could also check that $f_{yy}(1/3, -1/3) = -6(-1/3) = 2 > 0$.

Here is a picture:



1. Consider the scalar field

$$f(x, y, z) = xye^z.$$

a) Compute the gradient vector:

$$\begin{aligned}\nabla f &= \left\langle \frac{df}{dx}, \frac{df}{dy}, \frac{df}{dz} \right\rangle \\ &= \langle ye^z, xe^z, xye^z \rangle\end{aligned}$$

b) Compute the equation of the tangent plane to the surface $f(x, y, z) = 2$ at the point $(2, 1, 0)$.

$$[\text{Check: } f(2, 1, 0) = 2 \checkmark]$$

The normal vector is

$$\begin{aligned}\nabla f(2, 1, 0) &= \langle e^0, 2e^0, 2e^0 \rangle \\ &= \langle 1, 2, 2 \rangle\end{aligned}$$

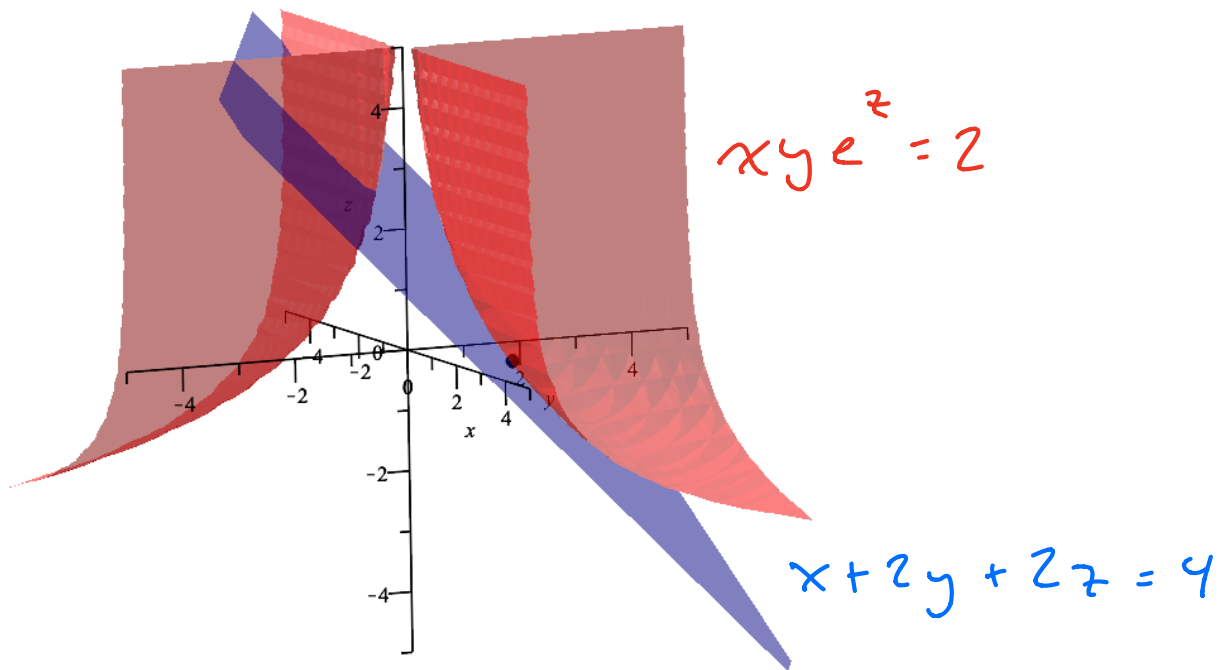
so the equation of the tangent plane is

$$1(x-2) + 2(y-1) + 2(z-0) = 0$$

$$x - 2 + 2y - 2 + 2z = 0$$

$$x + 2y + 2z = 4$$

Picture :



2. Consider the rectangular box with dimensions x, y, z and surface area

$$A = 2xy + 2xz + 2yz.$$

a) If x, y, z are functions of time then the chain rule gives

$$\frac{dA}{dt} = \frac{dA}{dx} \frac{dx}{dt} + \frac{dA}{dy} \frac{dy}{dt} + \frac{dA}{dz} \frac{dz}{dt}$$

First we compute

$$dA/dx = 2y + 2z + 0$$

$$dA/dy = 2x + 0 + 2z$$

$$dA/dz = 0 + 2x + 2y$$

Then we have

$$\frac{dA}{dt} = 2 \left[(y+z) \frac{dx}{dt} + (x+z) \frac{dy}{dt} + (x+y) \frac{dz}{dt} \right]$$

b) If the initial conditions are

$$\langle x(0), y(0), z(0) \rangle = \langle 2, 1, 2 \rangle$$

$$\langle x'(0), y'(0), z'(0) \rangle = \langle 3, -1, 1 \rangle$$

then the initial rate of change of the surface area is

$$\begin{aligned} A'(0) &= 2 \left[(1+2)(3) + (2+2)(-1) + (2+1)(1) \right] \\ &= 2 \left[9 - 4 + 3 \right] \\ &= 16 \quad \frac{(\text{unit of distance})^2}{\text{unit of time}} \end{aligned}$$

Remark : We don't have enough information to compute the rate of change $A'(t)$ for any time other than $t = 0$.